KARABAKH UNIVERSITY

FACULTY OF MATH & IT — FUNDAMENTALS OF PROGRAMMING

Course: Introduction to Python Programming
Assignment: Laboratory Assignment 1
Topic: Conditional Statements & Decision Making
Date: October 6, 2026
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1. Introduction & Objectives

This laboratory assignment focuses on applying conditional logic, decision-making algorithms, and branching statements in Python. Through three interactive applications, control flow structures (`if`, `elif`, `else`) and arithmetic operations are utilized to address practical computational problems.

2. Task 1: Body Mass Index (BMI) Calculator

Problem Description: The Body Mass Index (BMI) program evaluates a person's body mass relative to their height. The system prompts the user to input weight in kilograms and height in meters, calculates the index using the standard formula, and classifies the result into standard health categories.

Mathematical Model:
`BMI = weight / (height × height)`

BMI Range	Category
Below 18.5	Underweight
18.5 – 24.9	Normal Weight
25.0 – 29.9	Overweight
30.0 and above	Obese

Implementation Source Code

```
      a = 'y' while a == 'y': m = float(input('Enter weight: ')) h = float(input('Enter height: ')) BMI = m / (h ** 2) if BMI < 18.5: c = 'Underweight' elif BMI >= 18.5 and BMI < 25.0: c = 'Normal Weight' elif BMI >= 25.0 and BMI < 30.0: c = 'Overweight' else: c = 'Obese' print('BMI:', BMI) print('Category:', c)
```

3. Task 2: Chinese Zodiac Calculator

Problem Description: The Chinese Zodiac system follows a repeating 12-year cycle, with each year represented by a specific animal. The program computes the remainder of the birth year when divided by 12 using the modulus operator (%) and maps the outcome to the corresponding zodiac sign.

Remainder (year % 12)	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
Zodiac Animal	Monkey	Rooster	Dog	Pig	Rat	Ox	Tiger	Rabbit	Dragon	Snake	Horse	Sh

Implementation Source Code

```
     c = 'y' while c == 'y': a = int(input("Enter your year: ")) b = a % 12 if b == 0: z = "Monkey" elif b == 1: z = "Rooster" elif b == 2: z = "Dog" elif b == 3: z = "Pig" elif b == 4: z = "Rat" elif b == 5: z = "Ox" elif b == 6: z = "Tiger" elif b == 7: z = "Rabbit" elif b == 8: z = "Dragon" elif b == 9: z = "Snake" elif b == 10: z = "Horse" elif b == 11: z = "Sheep" print("Your Chinese Zodiac sign is:", z)
```

4. Task 3: Birthday Guessing Game

Problem Description: This program determines the user's birth day (from 1 to 31) through five binary choices. The algorithm displays five distinct sets of numbers representing binary bit-positions (1, 2, 4, 8, 16). When a user confirms that their birth day is in a specific set, the base value (first number) of that set is added to the accumulated sum.

Implementation Source Code

```
      a = 'y' while a == 'y': day = 0 print("Is your birthday in Set1?") print(" 1 3 5 7 9 11
13 15 17 19 21 23 25 27 29 31") answer = int(input("Enter 0 for No and 1 for Yes: ")) if
answer == 1: day += 1 print("\nIs your birthday in Set2?") print(" 2 3 6 7 10 11 14 15 18 19
22 23 26 27 30 31") answer = int(input("Enter 0 for No and 1 for Yes: ")) if answer == 1: day
+= 2 print("\nIs your birthday in Set3?") print(" 4 5 6 7 12 13 14 15 20 21 22 23 28 29 30
31") answer = int(input("Enter 0 for No and 1 for Yes: ")) if answer == 1: day += 4
print("\nIs your birthday in Set4?") print(" 8 9 10 11 12 13 14 15 24 25 26 27 28 29 30 31")
answer = int(input("Enter 0 for No and 1 for Yes: ")) if answer == 1: day += 8 print("\nIs
your birthday in Set5?") print("16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31") answer =
int(input("Enter 0 for No and 1 for Yes: ")) if answer == 1: day += 16 print("\nYour birthday
is:", day)
```

5. Conclusion & Verification

During this laboratory work, three separate Python programs implementing conditional logic were designed, tested, and verified:

- The **BMI Calculator** correctly computes floating-point mass-to-height ratios and accurately categorizes health indices.
- The **Chinese Zodiac Calculator** successfully demonstrates the usage of the modulus operator for cycle evaluation.
- The **Birthday Guessing Game** accurately models binary representation to determine values from 1 to 31 without direct input of the birth date.

> **Submission Status:** All tasks have been implemented according to university specifications, verified for computational accuracy, and prepared for GitHub repository submission.